Local review of GitHub issue 1
Analytic objections and rational certificate assembly

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6 November 2026 — local review, no external response sent

Abstract

We inspect the external review by GitHub user `wc72449`, compare its claims with the specified Kim–Scarinci version and the project's earlier records, and independently reassemble its supplied interval data using exact rational arithmetic. Both reported conservative positive margins survive this recomposition. The 21,143 cubature cells have not been reintegrated in this review.

1 What was inspected and what it establishes

Issue #1 was opened on 6 November 2026 and targets commit `714d038`. Its eight-page attachment and evidence ZIP were downloaded from the public issue, saved unchanged, and read locally. All 43 entries in its package hash manifest match; all 18 specified upstream files match the local files. The attached PDF also matches its ZIP copy. No supplied script was executed.

The original paper explicitly uses ordinary joint spectral calculus and states Proposition 6.15 for Weyl relations [1]. In its positive-curvature proof before (74), it asserts a cross strong commutation; after (75), it uses fixed-vector analytic continuation. These are the locations identified in the issue. They match the project's earlier flip-domain, weak-continuation and representation analyses. The external report therefore reinforces an existing objection; it does not concern the current Susskind memory calculation.

For the precise realization, use $\hbar = 1$ and

$$\mathcal{H} = L^2(\mathbb{R}^2, dx\,dy/(2\pi^2)), \quad P_x = -\pi i\partial_x, \quad P_y = -\pi i\partial_y. \tag{1}$$

The assignment $(\mathbf{x}, \mathbf{y}, \mathbf{x}', \mathbf{y}') = (X, Y, P_y, P_x)$ satisfies the regular Weyl hypotheses. With the paired function $F = F_1^1$ of the paper, put

$$A = F(X, Y), \quad B = F(P_y, P_x), \quad M = e^{iXY/\pi}, \quad D = AB - BM^{-1}BMA. \tag{2}$$

Differentiation on a core gives $M^{-1}P_yM = P_y + X$ and $M^{-1}P_xM = P_x + Y$; joint spectral calculus then identifies the sum factor with $M^{-1}BM$. The test vector $f = \sqrt{2}\exp[-(x^2 + y^2)/(2\pi)]$ has norm one. All factors in D are bounded unitaries. Intermediate chiral domains cannot exclude f from this final bounded identity.

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2 Analytic assessment

For $\tau > 0$ consider the normal operators $N = X - i\tau Y$ and $T = P_y + i\tau P_x$. Their formal commutator vanishes on the Schwartz core, whereas $[N, T^*] = 2\pi\tau I$. The decisive bounded witness follows from the Weyl relation:

$$e^{iX}e^{iP_x} = -e^{iP_x}e^{iX}, \qquad \|[e^{iX}, e^{iP_x}]u\| = 2\|u\|. \tag{3}$$

Since these unitaries are bounded Borel functions of N and T, the claimed cross strong commutation is false in this realization. This is an analytic conclusion independent of the interval certificate. It does not by itself disprove every identity for particular chiral functions on a specially chosen domain.

The first pole $z_0(h) = -\pi i(1 + h)$, with $h = a + ib$, $b > 0$, has real spectral preimage

$$x_h = \frac{\pi(a^2 + b^2 + a)}{b}, \qquad y_h = -\frac{\pi(1 + a)}{b}. \tag{4}$$

At $h = i$ this is $(\pi, -\pi)$, with residue -1 in the compact product normalization. A Gaussian nonzero there is outside the single chiral multiplication domain: its squared norm contains a nonzero multiple of $\int_0^\epsilon dr/r$. The pole map has positive Jacobian $\pi^2((a + 1)^2 + b^2)/b^3$, so the fixed-domain obstruction persists over an open parameter set. This supports the issue's objection to literal fixed-vector holomorphy. A different contour prescription would require an endpoint identification with the stated spectral operator. The coefficient in (75) also needs the factor i: $e^{\pi ih}$ becomes $e^{-\pi\tau}$ at $h = i\tau$.

3 Why the 9,890 audit failures require two separate verdicts

The external saved audit has `passed=false`. Inspection of all its failure strings finds only the following categories:

Overlap	changed bound endpoints
Gaussian	255
Left step 1	256
Left step 2	3016
Right step 3	2969
Right step 5	3394
Total	9890

The local audit's line 74 compares the new bound interval with a saved singleton upper endpoint. Such overlap is an exact regression condition; two valid conservative bounds need not satisfy it. The external portability summary reports no changed cell types. This inspection verifies what the stored failure report says, not the origin of its platform-dependent differences. We retain the failed regression status and assess the rebuilt certificate separately.

4 Independent endpoint recomposition

The local driver `review_issue1_evidence.py` reads the ZIP as data. Each binary64 Hermite coefficient is lifted to its exact dyadic rational. Norms and inner products are then summed in `Fraction` arithmetic. For positive rational q, use

$$k = \left\lfloor \sqrt{\lfloor 2^{640} q \rfloor} \right\rfloor, \qquad 2^{-320} k \le \sqrt{q} \le 2^{-320}(k+1). \tag{5}$$

The integer square root and rational inequalities certify these enclosures without using the supplied assembly code or Arb.

For each of the five nonlinear overlaps we recompute

$$e_j^2 \le \|v_j\|^2 + \|w_j\|^2 - 2(I_{j,\mathrm{low}} - E_{j,\mathrm{up}} - T_{j,\mathrm{up}} - P_{j,\mathrm{up}}), \tag{6}$$

using the saved integral, cubature-error, spatial-tail and product-tail endpoints. The two chirp bounds use their saved overlap endpoints. All seven saved upper bounds cover the corresponding independently reconstructed squared-error upper values. We nevertheless use our own square-root upper bounds for the final recomposition.

Writing l, r for the terminal trial vectors and E for the seven step bounds, the independently checked inequalities are

$$\|Df\| \ge \|l - r\| - E > 0.04489650872, \tag{7}$$

$$\inf_{\theta} \|ABf - e^{i\theta} BM^{-1}BMAf\| \ge \sqrt{\|l\|^2 + \|r\|^2 - 2|\langle r, l\rangle|} - E > 0.04489650465. \tag{8}$$

These conclusions are conditional on the supplied overlap enclosures and their analytic error estimates being correct. The local calculation verifies the final rational assembly, not fresh integration of every cell, and is not formal verification of the numerical libraries. The original local certificate gave a margin approximately 0.04489650872660254; the external recomposition agrees at the reported conservative threshold.

Research decision. Treat the issue as substantive corroboration with a useful phase test and a reproducibility diagnostic. Preserve the old evidence; do not relabel its failed external replay as passed. Full independent cubature and the cause of the endpoint changes remain follow-up checks. No external reply or author contact has been sent. Continue the Susskind main line from the completed two-return reduction to the finite-memory products; the positive-curvature construction in the older branch remains unrepaired.

References

[1] H. Kim and C. Scarinci, *A quantization of moduli spaces of 3-dimensional gravity*, arXiv:2112.13329v2, Secs. 6.3–6.5, especially the proof around (74)–(75) and Prop. 6.15.

[2] GitHub user `wc72449`, issue 1, 6 November 2026. External eight-page technical attachment and 43-file evidence manifest, archived locally with SHA-256 hashes.